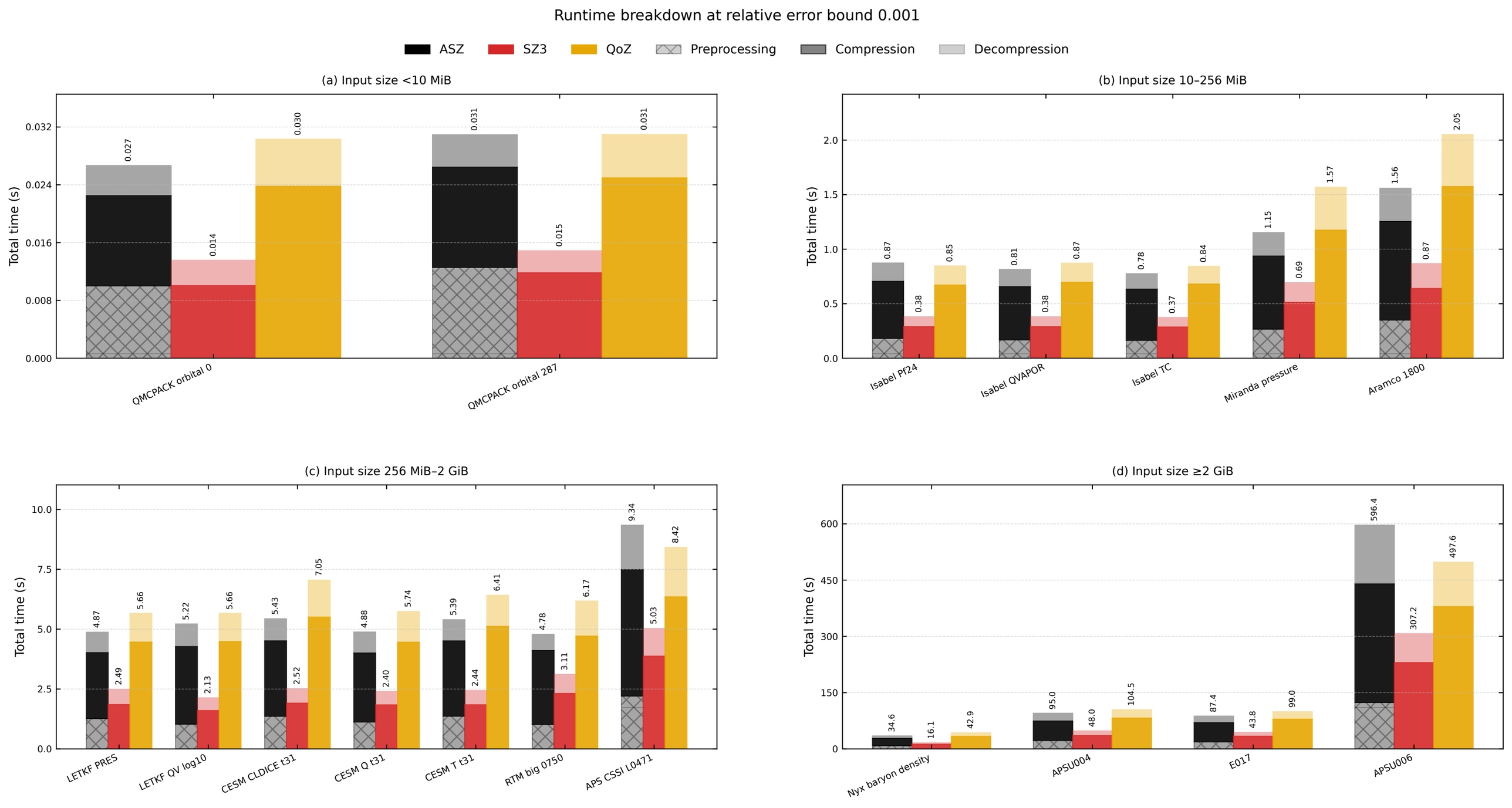
Runtime breakdown at relative error bound 0.001
ASZ
SZ3
QoZ
Preprocessing
Compression
Decompression
(a) Input size <10 MiB
Total time (s)
0.000
0.008
0.016
0.024
0.032
0.027
0.014
0.030
0.031
0.015
0.031
QMCPACK orbital 0
QMCPACK orbital 287
(b) Input size 10–256 MiB
Total time (s)
0.0
0.5
1.0
1.5
2.0
0.87
0.38
0.85
0.81
0.38
0.87
0.78
0.37
0.84
1.15
0.69
1.57
1.56
0.87
2.05
Isabel Pf24
Isabel QVAPOR
Isabel TC
Miranda pressure
Aramco 1800
(c) Input size 256 MiB–2 GiB
Total time (s)
0.0
2.5
5.0
7.5
10.0
4.87
2.49
5.66
5.22
2.13
5.66
5.43
2.52
7.05
4.88
2.40
5.74
5.39
2.44
6.41
4.78
3.11
6.17
9.34
5.03
8.42
LETKF PRES
LETKF QV log10
CESM CLDICE t31
CESM Q t31
CESM T t31
RTM big 0750
APS CSSI L0471
(d) Input size ≥2 GiB
Total time (s)
0
150
300
450
600
34.6
16.1
42.9
95.0
48.0
104.5
87.4
43.8
99.0
596.4
307.2
497.6
Nyx baryon density
APSU004
E017
APSU006